

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**B. Tech. EE Sem-VIII Theory Examination****(April-May, 2018)****EE 406/406.01 ELECTRICAL MACHINE DESIGN- II****Date: 30/04/2018, Monday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Figures to the right indicate full marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q - 1 Do as directed.****[4]**

- i). What types of slots are preferred for the induction motor?
- ii). What is slot space factor?
- iii). Which part of induction motor has the maximum flux density? What is the maximum flux density in that part?
- iv). What happens if the air gap length is doubled?

OR

Q - 1 Which factor should be considered when estimating the Average flux density in air gap of induction motor? Explain in detail. **[4]**

Q – 2 (A) During the stator design of a 3 phase delta connected, 30 kW, 400 V, 6 pole, 50 Hz, squirrel cage induction motor, following information has been obtained. Gross length of stator = 0.17 m, Internal diameter of stator = 0.33 m, Number of stator slots = 45, Number of conductors per slot = 12. Efficiency=88%, p.f.=0.86 lagging. current density of bar= 6 A/ mm² , current density of end-ring= 6.5 A/ mm² , mean diameter of end ring=28 cm. Use copper bars which has resistivity $\rho = 0.021 \Omega^* \text{ mm}^2 / \text{m}$. Based on the above data, design a suitable cage rotor for the above motor. **[10]**

Q –2 (B) Determine the approximate diameter and length of the stator core, the number of stator slots and the number of conductors for a 11 kW, 400 V, 3 phase, 4 pole, 1425 rpm delta connected induction motor. Adopt a specific magnetic loading of 0.45 Wb/ m² and a specific electric loading of 23,000 A/m. Assume full load efficiency and power factor 0.85 and 0.88 respectively. The ratio of core length to pole pitch is 1. The stator employs a double layer winding. **[7]**

OR

Q –2 (B) Calculate (a) diameter (b) length (c) number of turns per phase (d) full load current and cross section of conductors and (e) total copper loss of stator of 3 phase, 120 kW, 2200 volt, 50 Hz, 750 rpm star connected slip ring induction motor has following particulars: $B_{av} = 0.48 \text{ Tesla}$, $a_c = 26000 \text{ A/m}$, efficiency = 92%, p.f. = 0.88, $L = 1.25\tau$, $K_w = 0.955$, current density = 5 A/ mm² , Mean length of stator conductor = 75 cm, $\rho = 0.021 \Omega^* \text{ mm}^2 / \text{m}$ **[7]**

- Q – 3(A)** A 30 kW, 400V, 50 Hz, 6 poles star connected slip ring induction motor has the following design data: Stator core diameter=0.4 m, efficiency=0.9, power factor=0.8, flux per pole=12.4 mWb. Determine (a) number of stator slots (b) number of rotor slots (c) number of rotor conductor (d) rotor conductor area (e) rotor slot dimensions. The rotor voltage at slip ring at the standstill is to be about 200V. Make number of rotor slots 3 slots per pole pair less than the number of stator slots. The current density in rotor conductors is 4 A/mm^2 , rotor slot space factor is 0.4 and the ratio of depth of slots to width of slot is about 6. The winding factor assumed as 0.96. Taking $q=3$ for stator. Rotor is star connected. [7]
- Q – 3 (B)** Which factor should be considered when estimating the length of the air gap of induction motor? Explain in detail with necessary diagrams. [7]

SECTION – II

- Q – 4 Do as directed.** [4]
- i). Salient pole machines are not suitable for high speed operations, why?
 - ii). Mention the uses of damper windings in a synchronous machine?
 - iii). What is short circuit ratio?
 - iv). How the value of SCR affects the design of alternator?

OR

- Q – 4** What are the factors to be considered for selection of armature slots in salient pole machine? Explain in detail. [4]
- Q – 5 (A)** A 1250 kVA, 3.3 kV, 50 Hz, 300 rpm, 3 phase star connected alternator has 180 slots and 5 conductors per slot having single layer winding with full pitch coils and one circuit per phase. Determine the specific electric and specific magnetic loadings if the air gap diameter of alternator is 2.2 m and axial length is 0.5 m. Assume phase spread of 60° . [7]
- Q – 5 (B)** Following information have been obtained during the design of stator frame of a 500 kVA, 6.6 kV, 12 pole 500 rpm, star connected, 3 phase salient pole alternator. Stator diameter at gap surface = 1.3 m, Gross length of stator core = 0.21 m, Gap flux per pole = 0.0404 wb, pole arc to pole pitch ratio=0.7, SCR=1.1, Air gap mmf is 80% of no load field mmf, $K_g=1.15$, length of pole is 1cm less than gross length of core, leakage factor of pole is 1.15, flux density in pole body is 1.5 Tesla, full load field mmf is 2 times armature mmf, depth of field coil=0.04 m, space factor of field coil=0.7, stacking factor of pole=0.95, Field winding dissipates 700 W/m^2 , height of pole shoe and insulation height wise=0.04 m, flux density in rotor core is 1.1 Tesla. Based upon these information's, design the field system of this alternator, giving the following details. (i) Length of air gap (ii) Diameter of rotor at gap surface. (iii) Peripheral speed (iv) Length, width and height of the pole, (v) Depth of the rotor core. [10]

OR

- Q – 5 (B)** Determine the ampere turns required by the pole of an engine driven salient pole alternator which has the following specifications and dimensions: 1875 kVA, 3.3 kV, 50 Hz, 3 phase, 250 rpm. Internal diameter of stator = 238 cm, Gross stator core length = 38 cm, Flux per pole = 0.068 weber, Length of air gap = 0.6 cm, Height of the pole = 15 cm, Width of the pole = 16 cm. Ampere turns for the stator teeth and stator core taken together may be assumed 15 % of the air gap ampere turns. Pole Arc to Pole Pitch ratio is 0.65. Air gap contraction factor=1.15. Net axial length of pole is 1.5 cm less than Gross stator core length. **[10]**

B, Wb/m^2	0.5	0.707	1.0	1.18	1.23	1.35
at,A/m	100	175	285	420	560	670

[7]

- Q –6 (A)** 1000 kVA, 3 phase, 3300 V, salient pole alternator has the following data: Air gap diameter 1.2 m, length of core 0.4 m, number of poles 10, armature ampere conductors per metre 30000, ratio of pole arc to pole pitch 0.68, stator slot pitch 26 mm, current density in damper bars 4 A/ mm^2 . Design a suitable damper winding for the machine.

- Q –6 (B)** Write short note on SCR.

[7]
